

D.A.N.G.D.U.T

Your Ultimate Wireless Network Guardian

Solusi Praktis Wireless Intrusion Detection System

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whoami

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I am a Cybersecurity Enthusiast and Red Teamer specializing in Network Security, automated hacking tool development, biometric authentication auditing, and **Embedded Systems Security** (including physical IoT pentesting and hardware research).

2025 - Speaker @ idsecconf Makasar

2025 - Speaker @ Merdeka Siber

2025 - Speaker @ Skill Issue

2025 - Speaker @ Redacted Corp

2026 - Speaker @ Cyberbreaker S3

2026 - Trainer @ Redacted Corp

2026 - Speaker @ revolusiber

Evolusi Pertahanan: Dari C.A.S.T ke D.A.N.G.D.U.T.

- Sebelum mikrokontroler canggih, WiFi kampus adalah hukum rimba. Netcut merajalela.
- Untuk melawan, saya membuat *Counter Attack & Scanning Toolkit* (C.A.S.T).
- Sebuah *script* terminal (era BackTrack Linux) untuk mendeteksi pelaku ARP Spoofing dan langsung "memberi pelajaran" (DOS balik).
- **Dulu:** Butuh laptop tebal & skrip garang.
Sekarang: Ancaman berevolusi, alat kita menyusut jadi sensor saku pasif.



The Layer 2 Blindspot

Ilusi Zero Trust

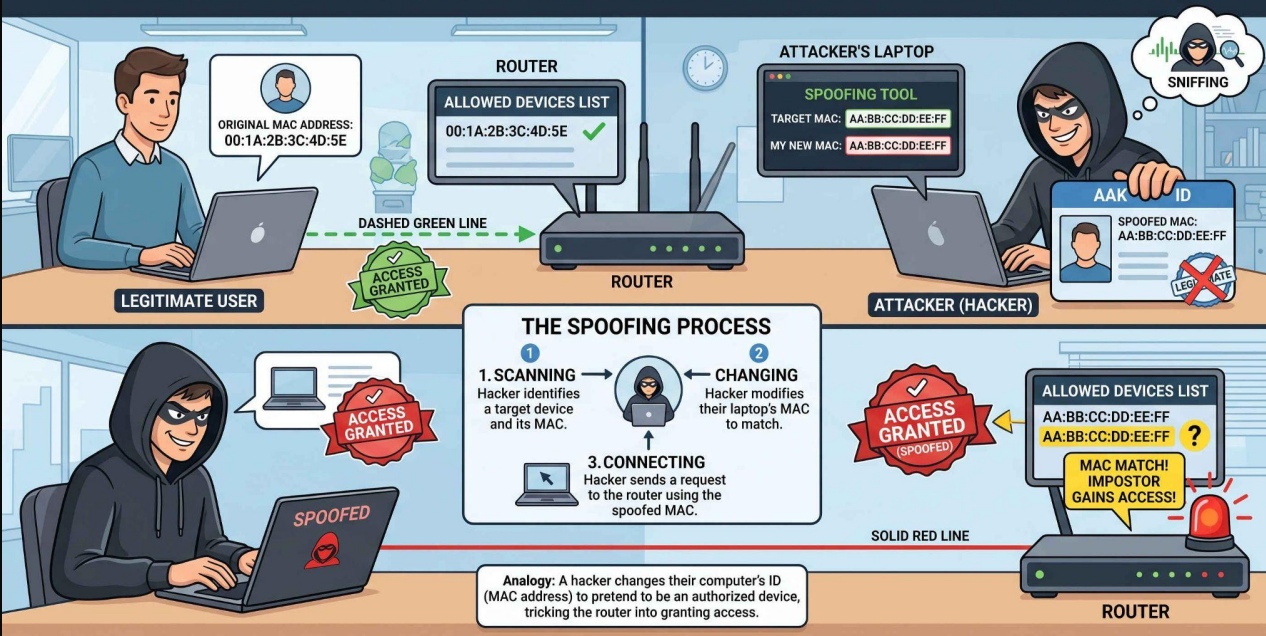
Banyak korporat/gov/coffee shop/mall/hotel menghabiskan anggaran besar untuk enkripsi *End-to-End*, namun abai jaringan wireless.

- ▶ Cacat Historis 802.11: *Management frames* dikirim dalam format *plaintext*.
- ▶ Ketidadaan integritas kriptografis memungkinkan siapa saja memalsukan kendali jaringan.
- ▶ Lapisan fisiknya sedang terbakar, dan banyak sistem keamanan *enterprise* tidak menyadarinya.



Validity vs Authenticity

MAC SPOOFING ANALOGY: IMPOSTOR ON THE NETWORK



- ▶ Ini bukan DoS berbasis *resource exhaustion*, melainkan eksploitasi logis protokol yang minim komputasi.
- ▶ Penyerang menangkap MAC Klien dan AP, lalu memalsukan *frame* Subtipe 12 dan 10 (Deauth/Disassoc).
- ▶ **Spoofing Dua Arah:** Menyamar sebagai AP ke klien, dan sebagai Klien ke AP.
- ▶ **Dampak:** Pencurian *handshake* & pembajakan **Evil Twin**.

Why Wi-Fi is Easy to Disrupt: Validity vs. Authenticity

Validity

(The Form of the Letter)

Checking the Physical Format

It only asks:

Is the **letterhead** official and is the **grammar** correct?

100% Protocol Compliance



Hacker "Deauth" commands perfectly follow IEEE 802.11 standards, making them appear "Valid."

Authenticity

(The Identity of the Sender)

Verifying the "ID Card"

It asks:

Is the person delivering this letter **actually** the authorized manager?



The Fatal Flaw of Blind Trust



Legacy Wi-Fi accepts packets from anyone claiming to be the Router (MAC Spoofing).

Minimal Effort for Hackers



Hackers use cheap tools to spoof identities because the system lacks cryptographic verification.

Memahami Ancaman: Pendekatan P.I.R



Problem

Ketiadaan kriptografi di layer manajemen.

Wi-Fi ibarat pos tanpa stempel. Siapapun bisa bikin 'surat pemecatan' palsu atas nama Bos (Router) ke Karyawan (HP/Laptop).



Impact

DoS & Intersepsi Evil Twin.

Jaringan Wireless mati serentak. Saat panik, karyawan nyasar ke Wi-Fi palsu kembaran (Evil Twin). Operasional lumpuh & password dicuri.



Recommendation

Wajibkan WPA3 & deploy Air-Gapped WIDS.

Jangka pendek, pasang alarm pengawas (DANGDUT). Alat ini teriak saat ada surat palsu, tim IT bisa langsung ringkus pelakunya.

Problem : Terjangkaunya Harga Alat Pemutus Wireless

Data baru dari API		Unduh CSV	Urutkan: Default	Termurah	Termahal
	PCB Nethercap WEMOS D1 ESP8266 Tokopedia			Rp 25.000	
	NodemCU ESP8266 NETHERCAP 2.7 — siap pakai, deauther & evil twin Shopee			Rp 58.000	
	Mini OLED Wifi Nethercap Siap Pakai - Deauther & Evil twin, Mudah Digunakan Shopee			Rp 68.000	
	NodemCu Nethercap 2.7 Full Version (Bukan Trial) Type C, Alat siap Pakai Tokopedia			Rp 69.900	
	KODE M32A Mini Oled Nethercap 27 Full Versi Bukan Trial Type C siap Pakai shopee.co.id			Rp 101.000	
	Mini oled nethercap terbaru siap pakai full version shopee.co.id			Rp 139.000	
	NETHERCAP & DeautherX shopee.co.id			Rp 140.000	
	Mini oled wifi nethercap siap pakai - MESIN+BOX tokopedia.com			Rp 159.000	

	Nethercap & Deauther versi terbaru dan permanen - Deauther X, Tanpa baterai tokopedia.com	Rp 170.000
	Mini Oled Nethercap shopee.co.id	Rp 185.000
	Nethercap mini oled oryinal Lazada Indonesia	Rp 199.000
	Mini oled Wifi nethercap siap pakai - Mesin + LCD+box tokopedia.com	Rp 230.100
	Nethercap PRO v3.6.3 lengkap tombol navigasi tokopedia.com	Rp 250.000
	TOP JT" NodeMCU type C "Nethercap 2.6" Sudah PROGRAM - Lngkp Drat+LCD tokopedia.com	Rp 324.470
	NETHERCAP & NovaX Dual-Band Bw16 support 2.4 ghz & 5 ghz - Nethercap Trial, Tanpa Baterai tokopedia.com	Rp 385.000
	NETHERCAP & NovaX 2.4ghz & 5ghz shopee.co.id	Rp 395.000
	NETHERCAP & NovaX Dual-Band Bw16 support 2.4 ghz & 5 ghz - Nethercap Pro, Dengan batrai tokopedia.com	Rp 485.000

Impact : Terjangkaunya Harga Alat Pemutus Wireless



A 44-year-old man from Australia faces lengthy jail time for creating a so-called evil twin WiFi network during a domestic flight.

#jail #prison #WiFi #Australia

504 83 148

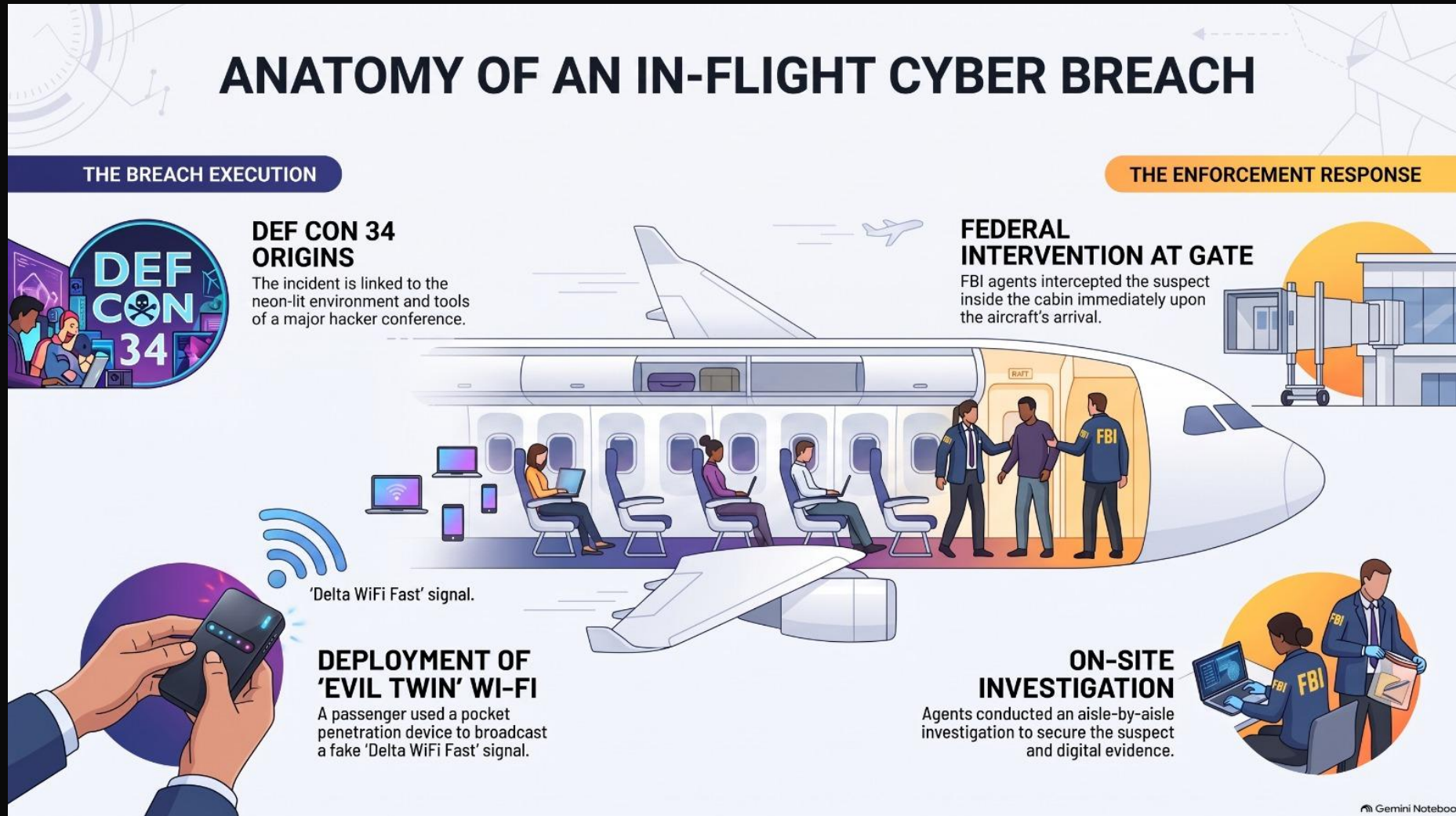
In April 2024, the Australian Federal Police (AFP) launched an investigation into the man after one of the flight's crew members had identified [a suspicious WiFi network mimicking a legitimate access point](#). This is also known as an evil twin network.

Police officers searched the man's hand luggage when he arrived at Perth Airport, seizing a portable wireless access device, laptop, and mobile phone.

Curious what others think about this story? Contribute your thoughts to the debate below.

<https://cybernews.com/security/man-sentenced-7-years-prison-evil-twin-wifi-network-flight/>

Impact :



<https://www.tomshardware.com/tech-industry/cyber-security/passenger-returning-from-def-con-34-spoofs-delta-wi-fi-network-while-in-flight-using-pentest-tool-pilots-tell-ground-crew-to-alert-corporate-security-after-attendee-from-hacking-conference-brings-the-party-to-the-sky>

Enterprise WIDS vs D.A.N.G.D.U.T.

- ▶ **Enterprise WIDS:** Berbiaya tinggi, infrastruktur yang kompleks, menciptakan *blindspot* bagi tim *mobile*.
- ▶ **DANGDUT WIDS:** Utilitas portabel, *low-cost*, mengubah ESP32 menjadi radar presisi.
- ▶ **Filosofi Air-Gapped:** Sniffer pasif (RX only). Mematuhi etika, tidak memancarkan *frame* bahaya.
- ▶ **Agilitas:** *Drop-and-monitor* lusinan sensor secara sembunyi.

Vendor / Produk	Karakteristik Fitur WIDS / WIPS
Aruba Networks (HPE)	Menggunakan arsitektur HPE Aruba Networking WIDS/WIPS yang sangat andal dalam mengklasifikasikan perangkat asing dan mendeteksi anomali trafik.
Cisco / Cisco Meraki	Meraki memiliki fitur khusus bernama Air Marshal yang berfungsi sebagai WIDS/WIPS bawaan untuk mendeteksi serta memblokir ancaman nirkabel secara konstan.
Arista Networks	Memiliki teknologi <i>Marker Packet</i> yang dipatenkan pada Arista WIPS untuk mengeliminasi <i>false alarms</i> saat mendeteksi Rogue AP.
Ruijie Enterprise	Menyediakan opsi WIDS pada lini produk Access Point mereka (seperti seri Ruijie RG-MAP920) yang mencakup <i>Rogue AP detection</i> , isolasi pengguna, hingga <i>containment</i> .
EnGenius	Menghadirkan solusi keamanan nirkabel cerdas lewat AirGuard WIPS/WIDS untuk melindungi ekosistem WLAN dari perangkat jahat.
Fortinet	Mengintegrasikan fitur WIDS/WIPS melalui FortiAP yang dikelola langsung di bawah <i>firewall FortiGate</i> sebagai pusat kendali keamanannya.

Ekosistem Hardware

Karakteristik dan Spesifikasi Teknis	ESP32 Type C V1 (30-Pin)	ESP32-S2 (Flipper Dev Board)	ESP8266 / Wemos D1 Mini
Arsitektur Silikon CPU	Dual-core Xtensa LX6	Single-core Xtensa LX7	Single-core Tensilica L106
Kecepatan Komputasi (Clock)	Hingga 240 MHz	Hingga 240 MHz	80 MHz hingga 160 MHz
Konverter USB Serial	CH340 / CP2102 (Eksternal)	Native USB CDC	CH340 / CP2102 (Eksternal)
API Promiscuous Dasar	esp_wifi_set_promiscuous_rx_cb	esp_wifi_set_promiscuous_rx_cb	wifi_set_promiscuous_rx_cb
Logika Kelistrikan Indikator LED	Aktif-Tinggi (GPIO 2)	Aktif-Rendah (GPIO 15)	Bervariasi (Umumnya Aktif-Rendah)
Integrasi Host Native UART	Membutuhkan modifikasi kelistrikan	Plug-and-Play (Koprosesor)	Membutuhkan modifikasi kelistrikan

Hardware Filter & Channel Hopping

Hardware Filter Mask

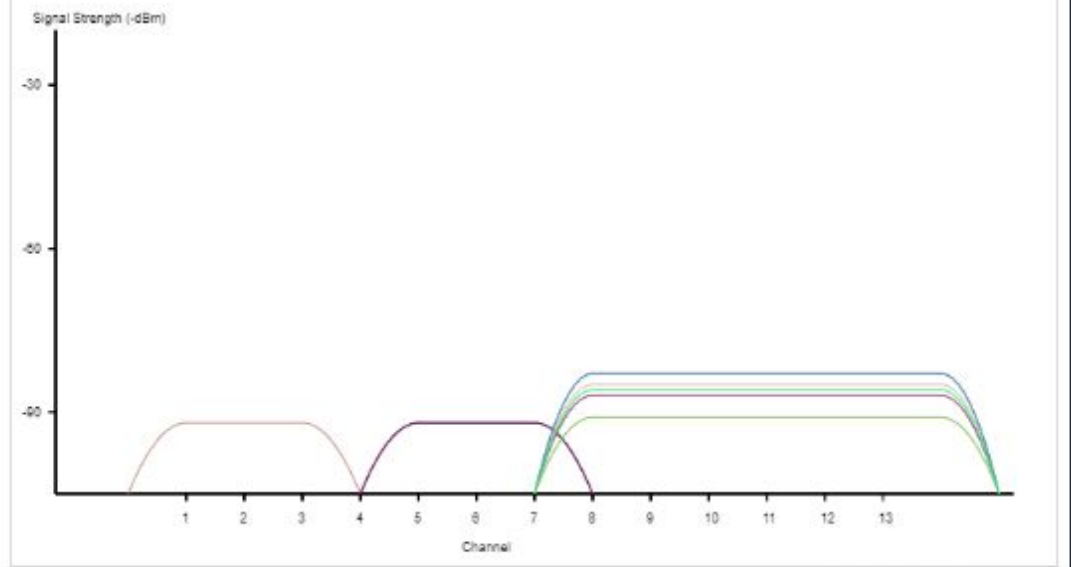
Membuang paket data normal di level *baseband* agar CPU murni mengurus *management frames*.

```
WIFI_PROMIS_FILTER_MASK_MGMT
```

Sliding Window Heuristic

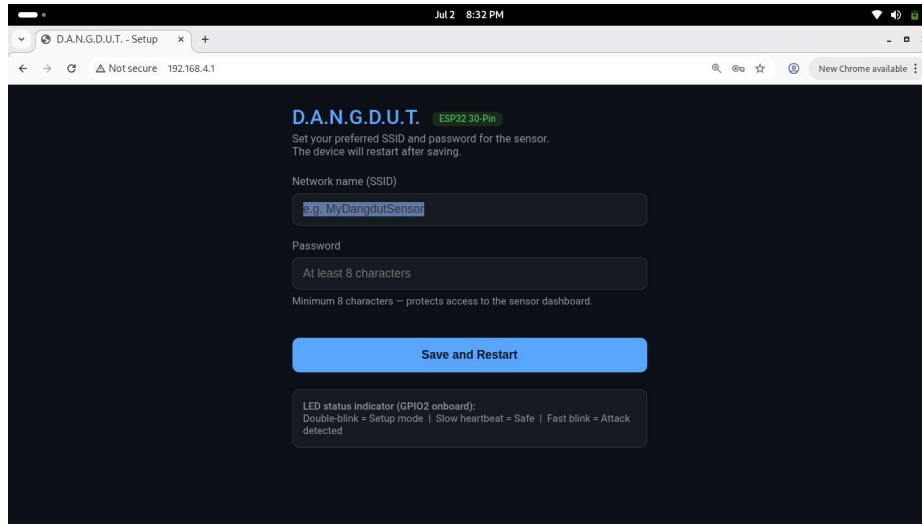
Mengamati jeda 5 detik. Jika ada >8 paket diskoneksi, status jadi ancaman kritis.

The below chart shows the results of the latest scan of your local area from your gateway.



Melompat Saluran 1-13 tiap 4 detik mencegah Kebutaan Spektral.

Web UI - D.A.N.G.D.U.T



D.A.N.G.D.U.T. ESP32-52 / Flipper Devboard

Deauthentication Attack Notification & Guard Detection Utility Toolkit

"Wireless Guardian in Your Pocket."

By XsanLahci

● SYSTEM SECURE

TOTAL PACKETS
5752

PACKETS / SEC
0

ACTIVE CHANNEL
4

TIME	TARGET (STA)	SENDER (AP/SPPOOF)	RSSI
00:03:46	A0:76:4E:9F:97:43	E8:2A:44:F7:4D:2B	-35 dBm
00:02:44	FF:FF:FF:FF:FF:FF	70:42:D3:BF:02:BE	-31 dBm
00:02:44	FF:FF:FF:FF:FF:FF	EC:B9:70:23:7B:AE	-34 dBm
00:02:44	FF:FF:FF:FF:FF:FF	72:42:D3:8F:02:BE	-32 dBm
00:02:44	FF:FF:FF:FF:FF:FF	72:42:D3:8F:02:BE	-33 dBm
00:02:44	FF:FF:FF:FF:FF:FF	72:42:D3:8F:02:BE	-32 dBm
00:02:44	FF:FF:FF:FF:FF:FF	72:42:D3:8F:02:BE	-32 dBm
00:02:44	FF:FF:FF:FF:FF:FF	EE:B9:70:83:7B:AE	-32 dBm
00:02:44	FF:FF:FF:FF:FF:FF	EE:B9:70:83:7B:AE	-33 dBm
00:02:44	FF:FF:FF:FF:FF:FF	EE:B9:70:83:7B:AE	-32 dBm

The screenshot displays the D.A.N.G.D.U.T. interface. At the top, the browser address bar shows the URL 192.168.4.1. The header section includes the tool name "D.A.N.G.D.U.T." in large blue letters, followed by "ESP32-S2 / Flipper Devboard" in a smaller font. A red button labeled "ATTACK DETECTED" is visible on the right. Below the header, there are three summary cards: "TOTAL PACKETS" with a value of 5780, "PACKETS / SEC" with a value of 2, and "ACTIVE CHANNEL" with a value of 9. A table below these cards lists packet capture data with columns for TIME, TARGET (STA), SENDER (AP/SPOOF), and RSSI. The table contains 10 rows of data, showing timestamps from 00:05:46 to 00:05:48, target and sender MAC addresses in FF:FF:FF:FF:FF:FF format, and RSSI values ranging from -19 dBm to -32 dBm.

TIME	TARGET (STA)	SENDER (AP/SPOOF)	RSSI
00:05:48	FF:FF:FF:FF:FF:FF	72:42:D3:8F:02:BE	-32 dBm
00:05:48	FF:FF:FF:FF:FF:FF	72:42:D3:8F:02:BE	-38 dBm
00:05:48	FF:FF:FF:FF:FF:FF	72:42:D3:8F:02:BE	-29 dBm
00:05:47	FF:FF:FF:FF:FF:FF	72:42:D3:8F:02:BE	-24 dBm
00:05:47	FF:FF:FF:FF:FF:FF	72:42:D3:8F:02:BE	-20 dBm
00:05:47	FF:FF:FF:FF:FF:FF	72:42:D3:8F:02:BE	-22 dBm
00:05:47	FF:FF:FF:FF:FF:FF	72:42:D3:8F:02:BE	-22 dBm
00:05:46	FF:FF:FF:FF:FF:FF	72:42:D3:8F:02:BE	-19 dBm
00:05:46	FF:FF:FF:FF:FF:FF	72:42:D3:8F:02:BE	-20 dBm
00:05:46	FF:FF:FF:FF:FF:FF	72:42:D3:8F:02:BE	-21 dBm

"The Cool Factor": Bypass Captive Portal

- Membuka tablet untuk *login* ke portal web saat insiden terjadi sangat membuang waktu.
- Solusi: [DANGDUT.fap](#)
- Sinyal intrusi dikirim langsung via pin GPIO 13 & 14 (UART baud 115.200).
- Menampilkan peringatan *pop-up* instan di layar Flipper Zero tanpa butuh koneksi IP.



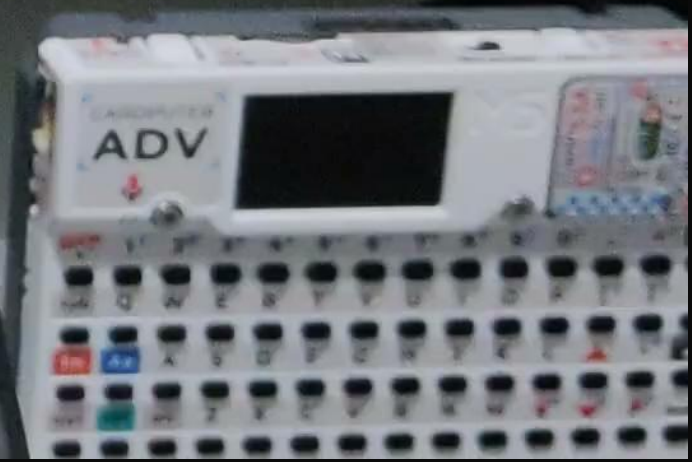
DEMO ?



**TURUNKAN
HARGA**

**WIFI
PINEAPPLE**







Hack
Danang Avan Maulana

Memb
Mengungkap manusi
Mengenali d



Community Edition VS PRO Edition

FEATURE	FREE Open Source	PRO Licensed
SETUP & BOOT LOADER		
Smart Captive Portal	✓	✓
Stealth AP / Hidden SSID	✓	✓
NVRAM Storage & Boot Animation	✓	✓
WIDS ENGINE		
Deauth Monitoring Start / stop on demand	✓	✓
Beacon Flood Detection	✗	✓
Karma Attack Detection	✗	✓
Auto Channel Hopping	✗	✓

Community Edition VS PRO Edition

FOX HUNT MODE		
Dynamic Radar Animation Track down the attacker	X	✓
Geiger Counter Audio	X	✓
ON-DEVICE UI / UX		
Battery Status & Charging Indicator	X	✓
Dynamic UI Slider Smooth scrolling menu & Wi-Fi scanner	X	✓
Submenu Config Ghost Mode, Network Info, Factory Reset	X	✓
ntfy.sh Wi-Fi Integration Connect to Wi-Fi to send push notifications on attacks	X	✓

Community Edition VS PRO Edition

WEB DASHBOARD		
Responsive Themes & Live Status Pills	✓	✓
Async AJAX & CSRF Protection	✓	✓
OUI / Vendor Lookup	✗	✓

DEMO

[01] THREAT_DETECTION_ENGINE

Built to spot the attacks that
break Wi-Fi trust.



Deauth Monitoring

Continuously watches 802.11 management frames and raises an alert the moment forced-disconnect deauth packets are detected on nearby networks.



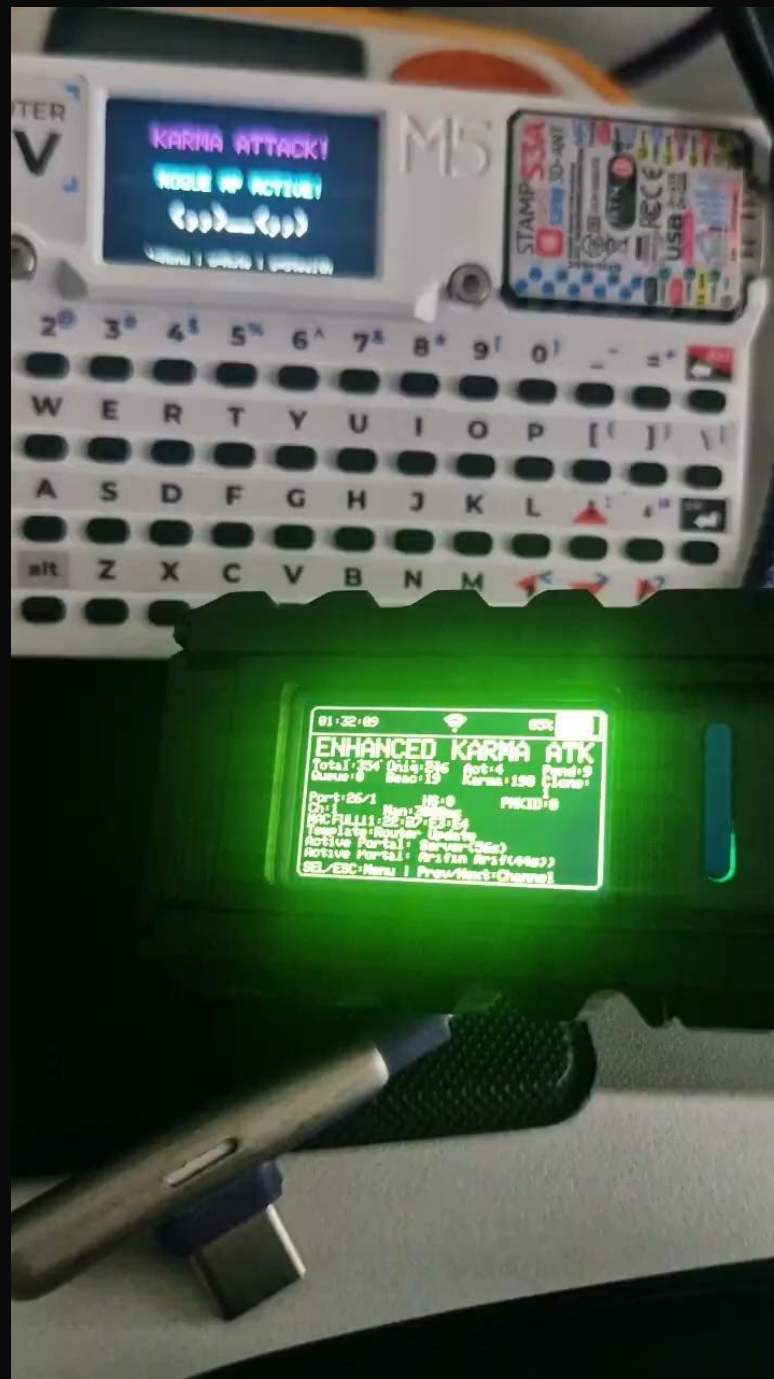
Karma Attack Detection PRO

Flags rogue access points that answer every probe request – the signature move of Karma-style evil twin traps.



Beacon Flood Detection PRO

Catches abnormal spikes in SSID beacon frames used to flood scanners and disrupt nearby wireless clients.



KARMA ATTACK!

MODE: HP ACTIVE!

Goodbye!!!

100% 100% 100%

01:32:00
ENHANCED KARMA ATK
Total: 304 Data: 500 Pot: 4 Spd: 9
Queue: 0 Recv: 10 Karma: 100 Score: 1
Port: 25/1 H2: 0 PWKID: 0
MAC: 11:22:33:44:55:66
ACTIVE Port: 25/1 H2: 0 PWKID: 0
ACTIVE Port: 25/1 H2: 0 PWKID: 0
SEL/ESC: Menu | Press/Menu: Charge



What is: Fox Hunting



Parker Molloy ✓

@parkermolloy.com



Batman in The Dark Knight (2008): I have to use mass surveillance tools in order to save Gotham. I struggle with this, but will have to learn to live with my decision.

Spider-Man in Brand New Day (2026): I'm scanning the faces of everyone in NYC and IDGAF



- Insiden WIDS terjadi di ruang fisik. Penyerang butuh kedekatan sinyal radio.
- DANGDUT mengekstrak nilai **RSSI** (Received Signal Strength Indicator) dari paket pelaku.
- Analis berjalan membawa alat, memantau perubahan angka desibel (misal: -87 dBm membesar jadi -45 dBm).
- Blue Team dapat melacak letak fisik *hacker* secara presisi berdasarkan kekuatan sinyal.



GAMBAR HANYA ILUSTRASI AI, TIDAK UNTUK DITIRU



PELAKU DEAUTH WIFI DI CAFE HABIS DIPUKULI

Notification via ntfy

9/20/26 00:42 

D.A.N.G.D.U.T WIDS

DEAUTH detected

9/20/26 00:39 

D.A.N.G.D.U.T WIDS

BEACON FLOOD detected

D.A.N.G.D.U.T

WIDS

Deauthentication Attack Notification & Guard Detection Utility Toolkit – your ultimate wireless network guardian for the M5Cardputer.

D.A.N.G.D.U.T turns your M5Cardputer into a pocket-sized Wireless Intrusion Detection System. It listens to the 2.4GHz spectrum around you and flags **Deauthentication**, **Karma**, and **Beacon Flood** attacks in real time – no laptop, no Wireshark, just a screen in your pocket.

[⚡ FLASH NOW VIA WEB SERIAL](#)[VIEW SPECS ↓](#)

Merespons Cepat di Era IoT Legacy

Menunggu migrasi universal ke PMF (Protected Management Frames) & WPA3 adalah strategi pasif. **DANGDUT WIDS** memberikan solusi taktis *compact* untuk memangkas MTTD (Mean Time to Detect) menjadi hitungan detik.

SHOUTZ & GREETZ



Alm. Pakde Hero / Lirva32 & Alm Pakde Randi/ Cybertank

My Family & Wife , Aurel666, Jos Ali Joe, sudo3rs, Petruknisme, Highphomet, Sal Scar,
DZ_AZ & All Member UTL





what will you build?

DESIGN THE FUTURE WITH ESP32 AND M5STACK



sebelum bertanya
pastikan pertanyaanmu
masuk akal

PAHAM 🖐️ 🤚

~ Ka GEM